

Multiplication in turing machine

A Turing Machine is a theoretical machine used in computer science to explain how computers solve problems step by step, which is a mathematical model that consists of three parts: Tape, Head and states

unary representation

The unary number system is the simplest way to represent natural numbers, A number is written by repeating a single symbol which is -1-as many times as the value requires

Example

1	→	1
2	→	1 1
3	→	1 1 1
4	→	1 1 1 1

The Multiplication Problem in Turing Machine

- One of the classic problems solved by a Turing Machine is multiplication.
- Given two numbers as input, the machine must compute their product and leave it on the tape as output.
- Since Turing Machines work best with simple symbols, we represent both numbers in unary.
- To separate the two operands on the tape, we place a 0 between them as a delimiter since 0 is not part of the unary alphabet
- The machine can always detect the boundary between the two numbers
- Multiplication in unary is performed using repeated addition, The machine copies the second number as many times as the first number indicates

The background is a solid dark blue color. Overlaid on this are several concentric, wavy lines in a lighter blue shade, creating a sense of motion and depth. These lines are arranged in a way that they appear to be emanating from or converging towards a central point, forming a complex, organic pattern. A subtle grid of thin, light blue lines is also visible, intersecting with the wavy lines to create a mesh-like effect.

Example

Multiply 2x3 using turing machine

The tape initially contains the two operands in unary, separated by a 0



Core Idea: For each '1' in the first number, copy the second number to the output

Replace a '1' in the first operand with X



q0 reads '1' > writes 'X', moves Right -> enters q1
The X marks this '1' as already processed.

Scan right past the '0' separator to reach the second operand.



► Head scanning RIGHT



q1 skips over the remaining 1s and the 0 separator without writing anything
head lands on first '1' of second operand.

For each '1' in the second operand, write a '1' at the output region



← original 2nd operand →

← 1st copy of result →



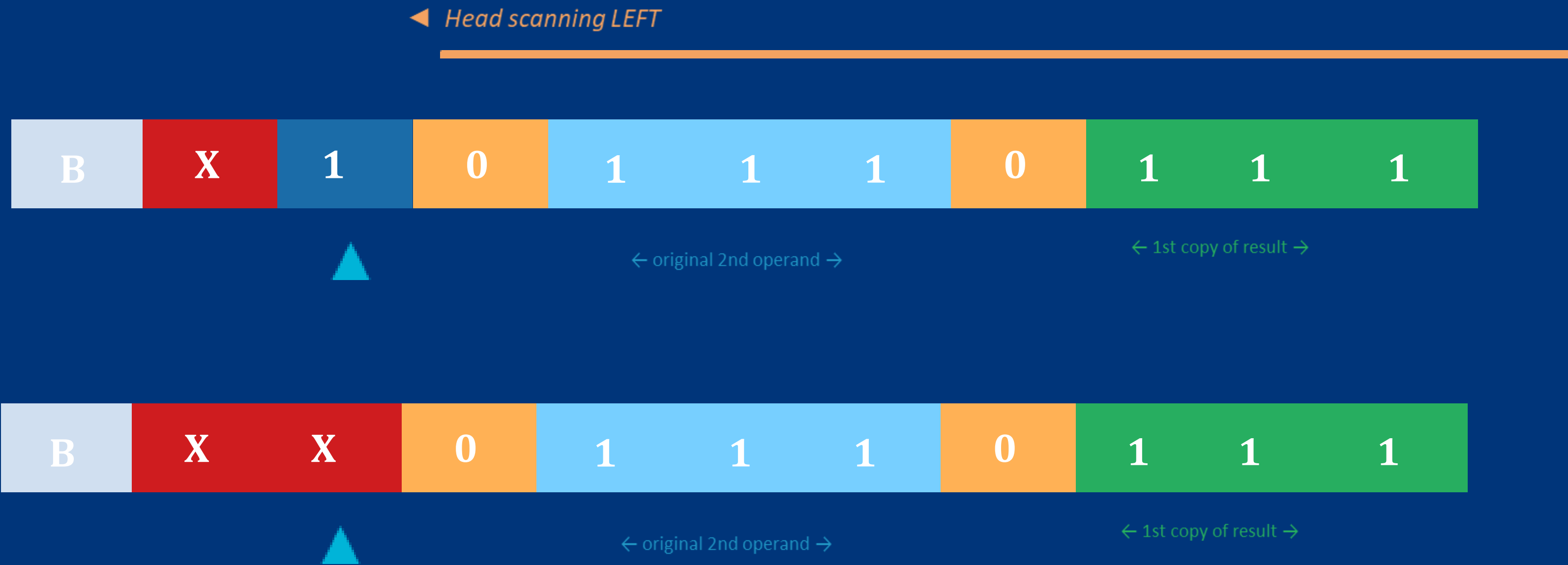
← original 2nd operand →

← 1st copy of result →



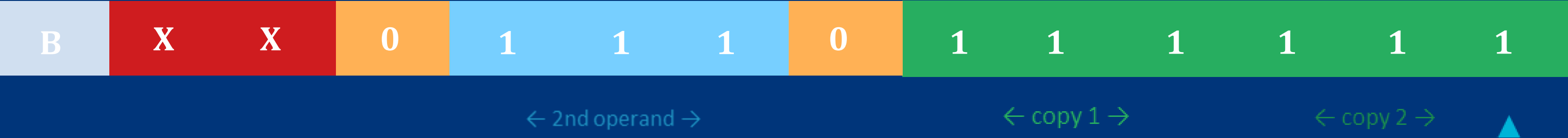
q2 writes one '1' per '1' found in the second operand, appending them right of the tape boundary.
Head now at first blank after the new 1s. head lands on first '1' of second operand.

Go back left. If an unmarked '1' remains in first operand, repeat steps 1-3.



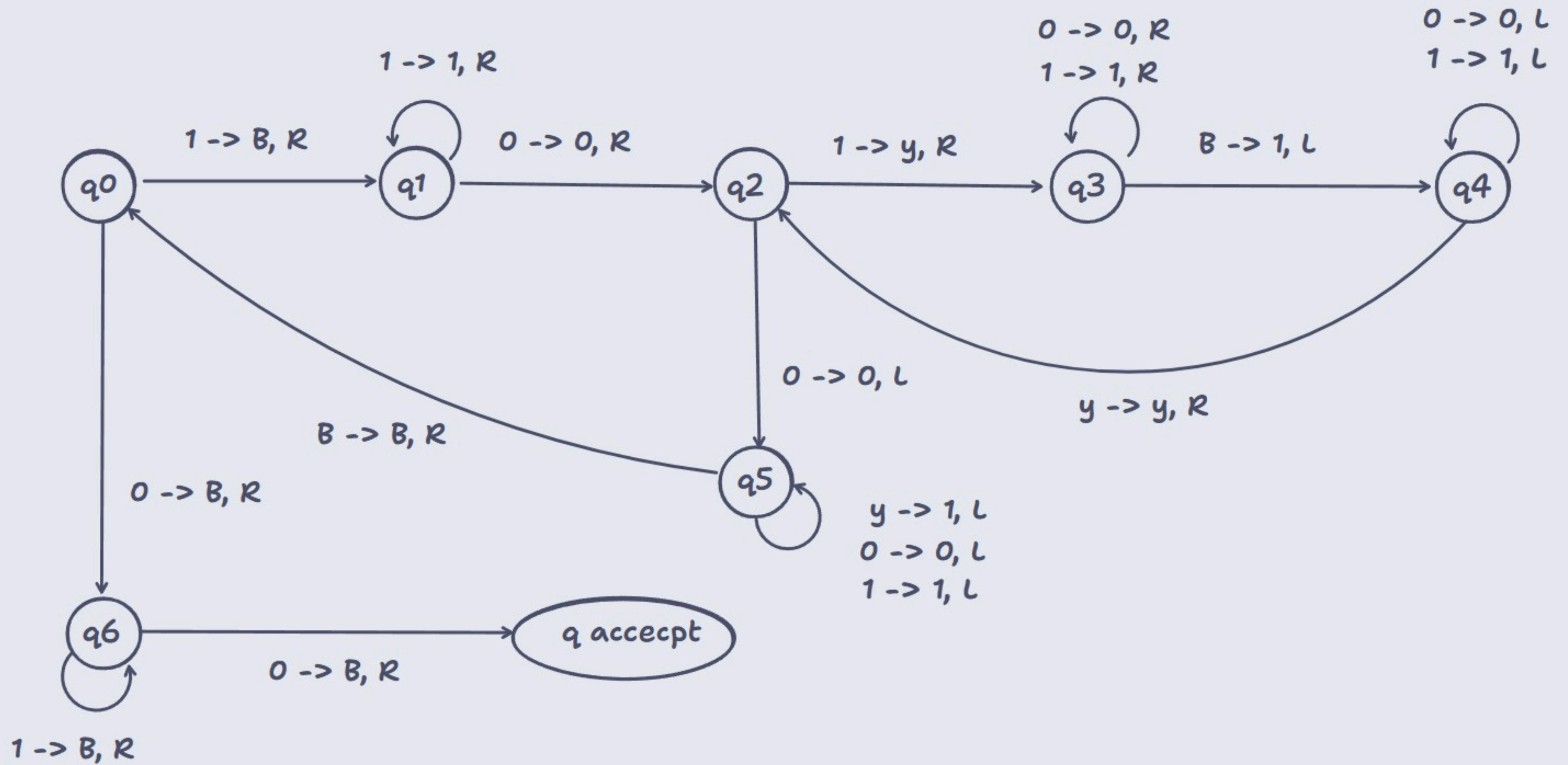
Both 1s of first operand are now marked X. The machine will copy '111' one more time to the output region.

Replace all X marks and separator '0' with blanks. Tape now holds the result.



Final Result





Algorithm steps

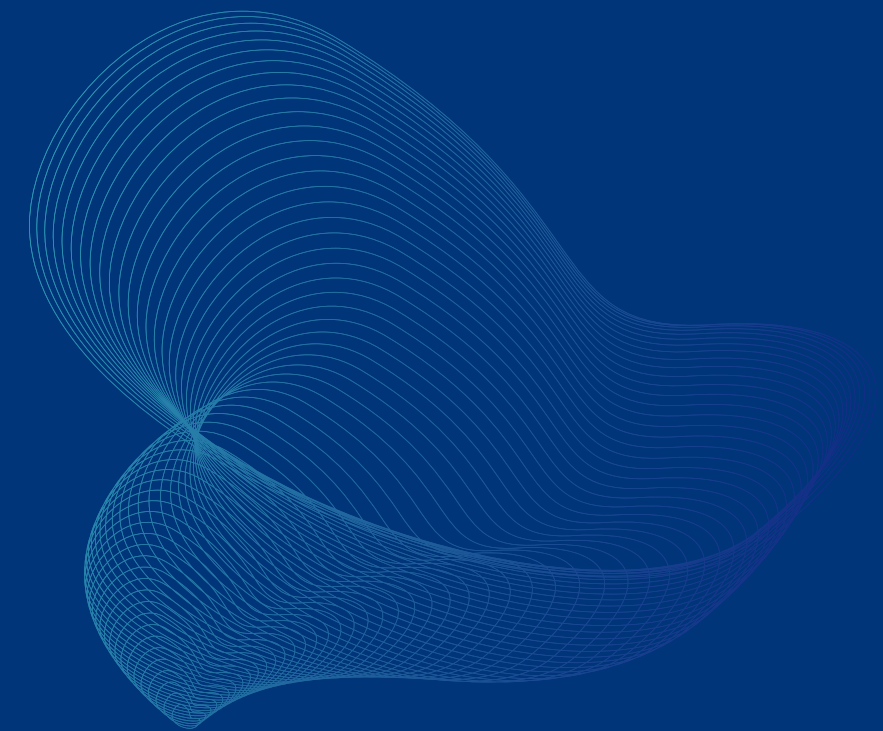
1 Mark a '1' In left group

2 Move past separator

3 Copy second operand

4 Return & repeat

5 Clean up & halt





Thank You